

Assignment 1: NumPy Array Surgery

The Dataset:

```
np.random.seed(42)
data_array = np.random.randint(1, 501, size=(10, 10))
```

The Question:

Using the `data\_array` provided:

1. Find the global mean of the entire array. Replace all elements in the array that are greater than this mean with \$0\$.
2. Calculate the sum of each column and the standard deviation of each row .
3. Slice the array to extract the center 4×4 sub-matrix (rows 3-6, columns 3-6) and convert it into a flat 1D array.

Assignment 2: Pandas Data Cleaning & Transformation

The Dataset:

```
raw_data = {
    'Product': ['Laptop', 'Mouse', 'Monitor', 'Keyboard', 'Laptop', 'Mouse', None],
    'Revenue': [1200, 25, 300, 75, 1150, None, 150],
    'Cost': [800, 10, 200, 40, 850, 15, 100],
    'Date': ['2025-01-10', '2025-02-15', '2025-03-20', '2025-10-05', '2025-11-12', '2025-12-25', '2025-06-01']
}
df = pd.DataFrame(raw_data)
```

The Question:

1. Handle Missing Values: Fill the missing `Revenue` with the median value of the column and the missing `Product` name with "Unknown".
2. Feature Engineering: Create a column `Profit` ($\text{Revenue} - \text{Cost}$) and a column `Margin\_Percentage` ($(\frac{\text{Profit}}{\text{Revenue}} \times 100\%)$).
3. Filtering: Convert the `Date` column to datetime objects. Filter the DataFrame to show only products sold in Q4 (October, November, December) that had a `Profit` greater than \$50\$.

Assignment 3: Grouping & Pivot Tables

The Dataset:

```
employee_data = {
    'Dept': ['IT', 'HR', 'IT', 'Sales', 'HR', 'IT', 'Sales', 'Sales'],
    'Status': ['FT', 'FT', 'Contract', 'FT', 'Contract', 'FT', 'Contract', 'FT'],
```

```
'Salary': [95000, 60000, 70000, 80000, 55000, 98000, 72000, 85000]
}
emp_df = pd.DataFrame(employee_data)
```

The Question:

1. Group the data by `Dept` and find the average salary and maximum salary for each department.
2. Create a Pivot Table that shows the count of employees, with `Dept` as the index (rows) and `Status` as the columns.
3. Sort the pivot table to identify which department has the highest number of `Contract` workers.